

Child Malnutrition Among Under-Five Children in Nigeria: A Systematic Review and Meta-analysis of Evidence from 2019–2024

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EXECUTIVE SUMMARY

Child malnutrition remains a critical public health challenge in Nigeria, particularly among children under five. This systematic review and meta-analysis synthesized evidence from studies published between 2019 and 2024 to estimate the prevalence of stunting, wasting, and underweight using WHO Child Growth Standards. Twenty-eight studies were included following a structured selection process. The pooled prevalence estimates were 35.43% for stunting, 15.84% for wasting, and 23.02% for underweight. Regional differences were observed, with pooled stunting prevalence substantially higher in Northern Nigeria (42.02%) than in Southern Nigeria (approximately 28%). The findings highlight a substantial and geographically uneven burden of undernutrition and support the need for context-specific nutrition strategies that address underlying socioeconomic, food-security, educational, and health-service factors.

BACKGROUND

Malnutrition in early childhood has implications for survival, development, educational outcomes, and future productivity. In Nigeria, stunting, wasting, and underweight remain important indicators of nutritional and health inequities. The burden is not distributed uniformly across populations or regions, making it important to synthesize recent evidence in a way that captures both the national picture and regional variation.

This review was designed to provide an updated synthesis of evidence on malnutrition among Nigerian children under five using a consistent framework based on WHO Child Growth Standards. By bringing together findings from multiple studies, the analysis sought to provide a clearer estimate of the burden and identify patterns that can inform public-health responses.

RESEARCH OBJECTIVE

The study aimed to systematically review and meta-analyse evidence published between 2019 and 2024 on the prevalence of stunting, wasting, and underweight among children aged 0–59 months in Nigeria, using WHO Child Growth Standards.

METHODS

Search strategy

A systematic review approach was used to identify and synthesize relevant evidence. Searches were conducted across PubMed, Scopus, Web of Science, Google Scholar, African Journals Online (AJOL), and other identified sources including WHO and UNICEF publications. Search terms included combinations of “malnutrition,” “under-nutrition,” “stunting,” “wasting,” “underweight,” “children under five,” “Nigeria,” and “prevalence.” Boolean operators (AND, OR) were used to refine the search. The search covered studies published from 2019 to 2024.

Eligibility and study selection

Studies were eligible when they were conducted in Nigeria, focused on children aged 0–59 months, reported quantitative prevalence data for at least one of stunting, wasting, or underweight using WHO Child Growth Standards, and met the specified publication and study-design criteria. Review articles, editorials, opinion pieces, studies conducted outside Nigeria, studies without relevant prevalence data, and studies using non-WHO definitions were excluded. The selection process followed PRISMA 2020 principles.

Data extraction and analysis

A total of 50 records were identified. Following duplicate removal, title and abstract screening, and full-text assessment against predefined criteria, 28 studies were included in the final synthesis. Data extracted from the selected studies were compiled in Microsoft Excel and imported into STATA for statistical analysis. Study quality was assessed using the Newcastle-Ottawa Scale, and quantitative synthesis was conducted for the major malnutrition indicators.

KEY FINDINGS

Pooled prevalence

- Stunting: 35.43% of children under five.
- Wasting: 15.84% of children under five.
- Underweight: 23.02% of children under five.

The review identified substantial variation across the included studies, reflecting differences in study populations, settings, and contextual factors. Statistical heterogeneity was high, indicating that the observed prevalence estimates were not uniform across studies and should be interpreted in light of population and contextual differences.

Regional variation

A notable regional pattern emerged for stunting. The pooled prevalence was 42.02% in Northern Nigeria compared with approximately 28% in Southern Nigeria. This difference points to important geographic and socioeconomic inequalities in child nutritional outcomes.

INTERPRETATION

The findings indicate that undernutrition remains a substantial public-health concern among Nigerian children under five. Stunting had the highest pooled prevalence among the three indicators examined, suggesting a particularly important burden of chronic undernutrition. Wasting and underweight also affected a considerable proportion of children.

The regional differences in stunting suggest that national estimates alone may mask important local variation. The observed disparities may be influenced by interacting socioeconomic, cultural, food-security, maternal-education, and health-service factors identified in the reviewed evidence. The high heterogeneity across studies also emphasizes that interventions should be informed by local context rather than assuming that a single approach will have the same effect across all populations.

PUBLIC HEALTH IMPLICATIONS

- Nutrition interventions should account for regional differences in the burden and determinants of child malnutrition.
- Strategies addressing both chronic and acute forms of undernutrition are needed rather than focusing on a single indicator.
- Nutrition policies should consider underlying socioeconomic conditions, food insecurity, maternal education, and access to health services.
- Continued surveillance and high-quality research are important for tracking nutritional outcomes and evaluating interventions.
- Evidence synthesis can help decision-makers interpret findings across individual studies and identify patterns that may not be visible from single studies.

CONCLUSION

This systematic review and meta-analysis synthesized evidence from 28 studies published between 2019 and 2024 and found substantial levels of stunting, wasting, and underweight among Nigerian children under five. The pooled estimates—35.43% for stunting, 15.84% for wasting, and 23.02% for underweight—demonstrate the continuing importance of child nutrition as a public-health priority. The higher prevalence of stunting observed in Northern Nigeria further highlights the importance of geographically and contextually responsive interventions.

RESEARCH SKILLS DEMONSTRATED

This project demonstrates experience in systematic literature searching, evidence screening and selection, application of predefined eligibility criteria, data extraction, quality assessment, quantitative synthesis, statistical interpretation, scientific writing, and evidence-based recommendations.

SELECTED REFERENCES

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Note: This writing sample is an edited professional presentation of the author's academic research project. The underlying study design, methods, findings, and terminology are retained from the original project.